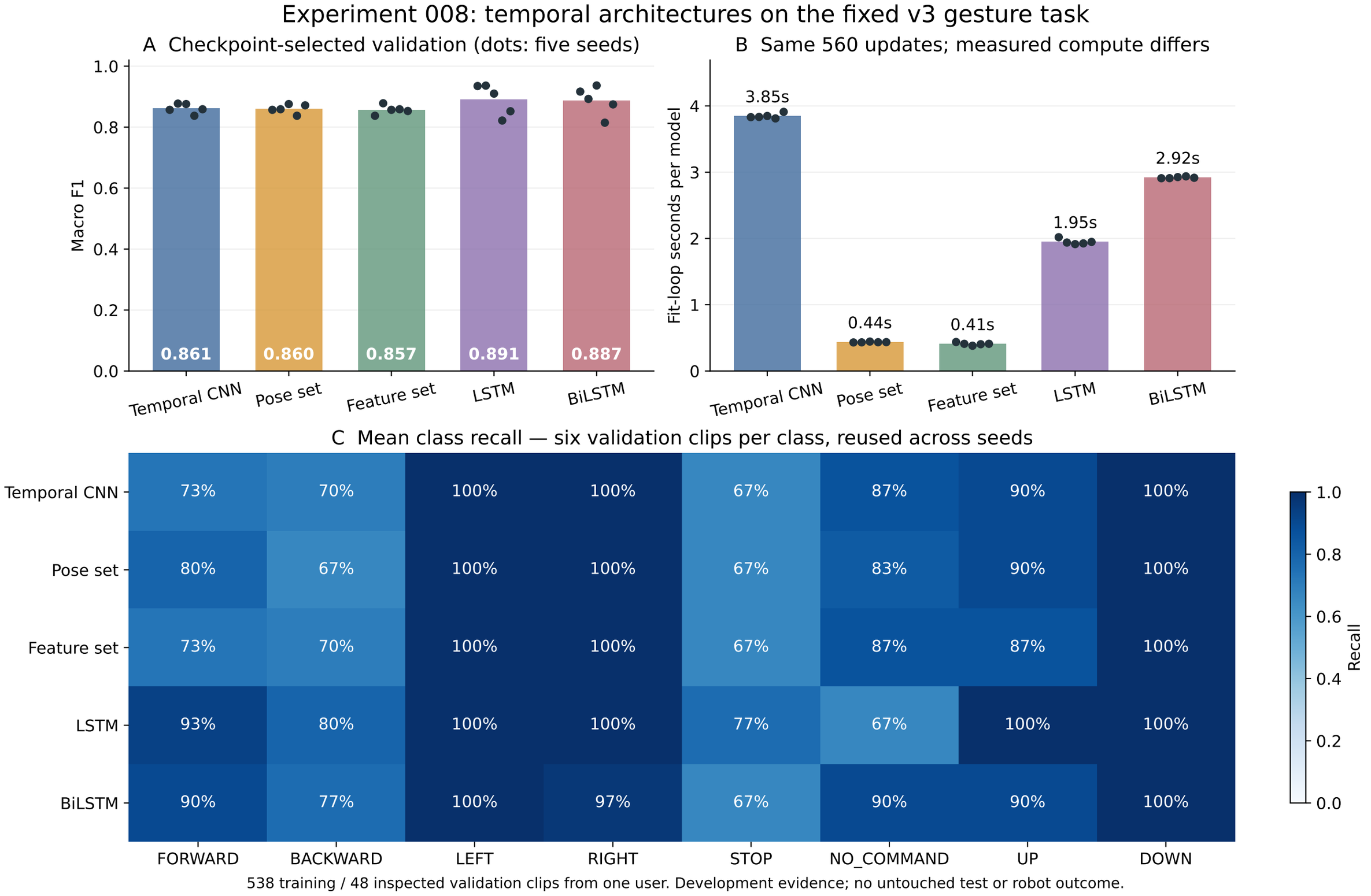
Experiment 008: temporal architectures on the fixed v3 gesture task
A Checkpoint-selected validation (dots: five seeds)
Macro F1
0.861
0.860
0.857
0.891
0.887
Temporal CNN
Pose set
Feature set
LSTM
BiLSTM
B Same 560 updates; measured compute differs
Fit-loop seconds per model
3.85s
0.44s
0.41s
1.95s
2.92s
C Mean class recall — six validation clips per class, reused across seeds
Temporal CNN 73% 70% 100% 100% 67% 87% 90% 100%
Pose set 80% 67% 100% 100% 67% 83% 90% 100%
Feature set 73% 70% 100% 100% 67% 87% 87% 100%
LSTM 93% 80% 100% 100% 77% 67% 100% 100%
BiLSTM 90% 77% 100% 97% 67% 90% 90% 100%
FORWARD
BACKWARD
LEFT
RIGHT
STOP
NO_COMMAND
UP
DOWN
Recall
538 training / 48 inspected validation clips from one user. Development evidence; no untouched test or robot outcome.